

# ernie

A daily visit that notices a bad day at home before it becomes a hospital day.

Christopher Pluntke · September 2026



# Margaret is 84.

Every morning she signs Ernie's book. This morning she didn't.

Hypoactive delirium, the most common kind, looks like nothing at all: quieter, slower, a little withdrawn. It has a cause you can treat in a day: an infection, dehydration, a new tablet.

**60–75%**

missed, even in hospital

**0%**

noticed at home, where it starts



[Start again](#)

Step 1 of 4: Signing in

**Thank you,  
Margaret.**

✓ **Done**

Lovely to see you. Here is your page in the book.



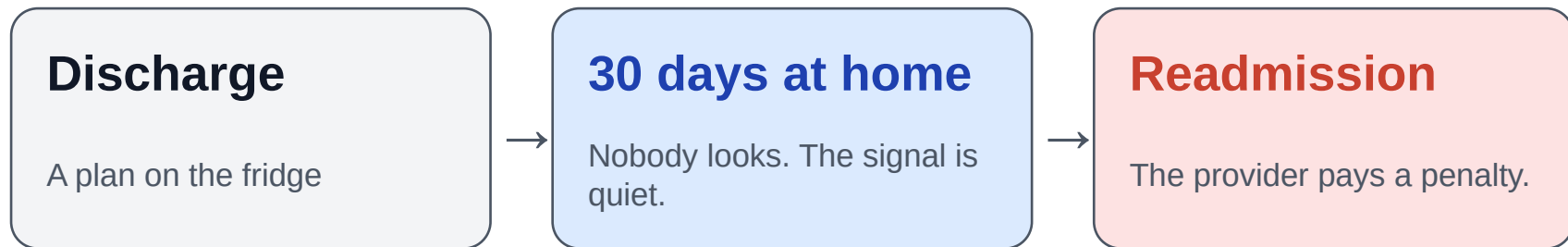
**Thursday 17 September**

**Carry on**

# The missing link is the 30 days at home.



Value-based care pays providers for outcomes. The outcome is decided at home, between visits. Nobody is there daily.



**\$36 B**

a year in 30-day readmissions

**2.0 M**

readmissions a year

**up to 3%**

of Medicare payments at risk  
(HRRP)

**\$8 k**

extra per delirium episode

# No tests. Just a visit.

Sign in, a few questions in her own words, a puzzle with Anna, goodbye. Every one of them is also a measurement.

Start again

Step 1 of 4: Signing in

**Good morning, Margaret.**

Sign my book to come in.



Any signature is fine. There is no wrong way.

Start again

Done signing

1 Sign the book

Start again

Step 2 of 4: A few questions

**Margaret, may I ask you a few things?**

There are 6 things today. It takes a few minutes, and you can stop whenever you like.

Answers are by tapping today.

Not today

Yes, go ahead

2 A few questions

Start again

Step 3 of 4: A puzzle

**Finish the picture**

Drag a piece onto the picture.  
Anna will help.

**11 pieces left · Anna is here**



Show me where this goes

Leave it for now

3 A puzzle, with Anna

Start again

Step 4 of 4: Goodbye

**Goodbye, Margaret.**

✓ **Done**  
That is everything for today.

Thank you for your company.  
I will be here again tomorrow morning.

Start again

4 Goodbye

# What the phone saw.

*“Margaret’s signature was slower, more hesitant and a different shape than usual today.”*

## Her own baseline

Five signings, median and spread per feature.  
No population norms; delirium is a change from your usual.

## Nine features

Time to start, total time, pauses, strokes, speed, smoothness, size, shape.

## Robust z

Today minus usual, over the spread. Two features past 2 is different from usual; three days is a trend.

## Who sees it

The family and the clinic. Margaret sees a thank-you and her page in the book.

Margaret's signature was slower, more hesitant and a different shape than usual today.

### SIGNING IN

Time to first stroke  
**418 ms**  
initiation

Total time  
**4.09 s**  
first touch to last lift

In-air time  
**1.07 s**  
between strokes

Strokes  
**3**  
pen lifts + 1

Mean speed  
**0.73 pad/s**  
path over drawing time

Smoothness  
**-15.95**  
log dimensionless jerk

Size  
**77% × 21%**  
of pad width

Date answer  
**correct**  
134 ms

| FEATURE              | TODAY         | USUAL         | ROBUST Z             |
|----------------------|---------------|---------------|----------------------|
| Time to first stroke | 418 ms        | 422 ms        | -0.1                 |
| <b>Total time</b>    | <b>4.09 s</b> | <b>3.06 s</b> | <b>+4.2 unusual</b>  |
| <b>In-air time</b>   | <b>1.07 s</b> | <b>268 ms</b> | <b>+37.5 unusual</b> |
| Strokes              | 3             | 3             | +0.0                 |

# Why delirium first.

Every instrument measures the same four things. None can see the fifth, because nobody re-tests hours later. A daily visit can.

## Attention

days backwards, search time in the puzzle

## Orientation

"What is today?" as a friend would ask

## Disorganised thinking

coherence of her own words, order in the puzzle

# 40%

of delirium is preventable with water, sleep, glasses, hearing aids and a phone call. A flag has a cheap answer.

## Arousal

did she open it, how fast did she start

## Fluctuation

morning vs evening, day vs her own baseline

*Delirium is a change from your own baseline. The home never had one.*



# The visit carries the clinic's questions.

A nurse writes the question in plain words. Ernie asks it every morning.

What does the scale say this morning?

Is it harder to breathe today than yesterday?

Can you say the days of the week backwards?

What have you been up to today?

*Refused at the desk: clinical words, instructions, more than six a day.*

## When something changes, explain the loop

### What changed

Her signature was slower and she paused longer.

### What it usually means

A change in attention can be the first sign of an infection or a new tablet.

### Who to call

The care line today. Not 911.

*Design, not yet built. Decision support, never a diagnosis; a human on every flag.*

# What's next

1

## Data leaves the phone

One process, one file: events, baselines, a family link.

2

## Three real people, three days

Does a finger signature hold a baseline? Does the doodle beat the check-in?

3

## The explanation layer

A flag opens what changed, what it means, who to call.

How it was built: nine design docs, a component kit that bakes in the 80+ criteria, a critic that audits every screen, and Claude for the research, the reading of replies, and the code.



*Synthetic people only. Not a medical device.*